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PROFESSIONAL SUMMARY

Final year BSc Computer Engineering undergraduate with strong experience in both AI/ML and software development. Skilled in designing and deploying intelligent applications that combine modern web technologies and advanced AI technologies. Proficient in Python, FastAPI, React, TensorFlow, LangChain, n8n and with practical experience in Generative AI, RAG systems, AI voice agentic systems and agentic Automation. Experienced in building scalable full stack applications using the NestJS and cloud platforms such as AWS and Firebase. Passionate about developing innovative, real world solutions that merge AI intelligence with robust software engineering to deliver impactful user experiences.

EDUCATION

B.Sc. (Hons) Computer Engineering, University of Jaffna (2022 – Present)

EXPERIENCE

- AI/ML + Software Engineer Intern, Idea8 Pvt Ltd, Kottawa (2025 Nov – 2026 Apr)
- Freelancer at upwork.com (2026 Jan - Present)

PROJECTS

AI Voice Agent Platform for Automated Outbound Calling

Tech: React, n8n, Twilio, Vapi, Deepgram, ElevenLabs, Pinecone, Supabase

- Built a production ready AI voice agent system for automated outbound calling, integrating LLM powered conversational logic with real time STT (Deepgram) and TTS (ElevenLabs) pipelines.
- Implemented long term conversational memory using Pinecone vector database for semantic retrieval of historical interactions, enabling context aware, personalized responses.
- Engineered agentic workflow orchestration via n8n, coordinating telephony (Twilio), LLM inference and automated follow up actions (SMS/email) based on conversation outcomes.
- Architected a multi tenant SaaS platform with a React campaign dashboard, supporting client onboarding, contact management and real time analytics.

AI Powered Insurance Consultation & Meeting Intelligence Platform

Tech: React, FastAPI, WebRTC, WebSockets, Gemini API, Pinecone, Deepgram, Supabase

- Designed an AI driven insurance consultation platform with real time video sessions (WebRTC) and a Deepgram STT pipeline for live meeting transcription and post-call storage.
- Developed a Retrieval Augmented Generation (RAG) system using Gemini and Pinecone, delivering context aware insurance recommendations grounded in internal policy knowledge bases.
- Engineered a real time AI assistant that streams LLM generated suggestions to agents during live meetings; built a meeting intelligence system with automated PDF export for compliance documentation.

Secure AI Medical Chatbot with Usage Control & Anti Abuse System

Tech: React, n8n, Supabase, Google Gemini API, FingerprintJS

- Developed a secure AI medical chatbot using Google Gemini API with session memory, supporting both guest and authenticated users via a real time conversational interface.
- Engineered a multi layered usage control system with per tier prompt limits (guest vs. authenticated), persistent tracking in Supabase and usage inheritance across sessions.
- Implemented anti abuse mechanisms combining device fingerprinting (FingerprintJS) and IP based rate limiting to prevent restriction bypass.

AI Powered Client Intake & Workflow Automation Chatbot

Tech: React, FastAPI, Supabase, OpenAI, Voice I/O, Google Cloud Run

- Built a full stack AI client intake system replacing form based onboarding with an LLM driven conversational pipeline supporting text and voice I/O (STT/TTS).
- Engineered a structured NLP intake pipeline to extract and convert natural language responses into review ready data with deterministic mission summaries.
- Integrated Monday.com API for end to end workflow automation and built audit logging for full compliance and traceability of client submissions.

Scalable HR Management Platform for Idea8 Pvt Ltd

Tech: NestJS (TypeScript), React, PostgreSQL, Prisma, WebSockets

- Designed a production grade HR platform with multi role access control (Admin, HR, Employee), JWT authentication and a modular NestJS backend with Prisma ORM.
- Implemented a real time notification system using WebSockets for instant HR event alerts and scheduled cron jobs for automated email notifications via Nodemailer.
- Built workflow driven leave and WFH approval pipelines with full auditability and deployed in AWS.

Multimodal AI System for Eye Disease Diagnosis (Research)

Tech: CNN (InceptionV3), BERT, ViT, RAG, FAISS, Google Gemini, PyTorch, TensorFlow, Streamlit

- Engineered a multimodal diagnostic system with three inference pipelines: CNN (InceptionV3) for eye image classification, BERT for clinical NLP reasoning and a joint fusion model combining image and text embeddings.
- Built a RAG pipeline using FAISS and Gemini for medical knowledge grounding and context aware clinical question answering.
- Implemented a chat driven diagnostic assistant with LLM generated explanations for predicted conditions, improving interpretability for non expert users.

AI Powered Email Management & Approval Automation

Tech: n8n, Gmail API, OpenAI API, AI Agents, Excel Dashboard

- Designed and implemented an AI driven email automation workflow that classifies incoming emails, filters spam, prioritizes business inquiries, and generates context aware professional reply drafts.
- Built a human in the loop approval system where AI generated responses are saved as Gmail drafts for review before automated sending, ensuring accuracy and operational safety.
- Developed an Excel based tracking and dashboard solution to monitor email status, priorities, approvals, and workflow performance, enabling end to end visibility of business communications.

SKILLS AND ABILITIES

- Programming Languages: Python, Java, C++
- SE Stack: MERN, NestJS
- ML and DL Frameworks: Scikit-learn, TensorFlow, Keras, Random Forest, Regression Models
- Generative AI & NLP: LLMs, LangChain, Transformers, RAG, Prompt Engineering, Gemini, Groq, GPT models
- Agentic AI & Automation: LangGraph, n8n
- Web & API Development: FastAPI, Streamlit, Flask, React, REST APIs
- Databases & Vector Databases: PostgreSQL, MongoDB, Supabase, MySQL, ChromaDB, Pinecone
- DevOps & Tools: Google Colab, Hugging Face, Vercel, Railway, LLMops, Docker, AWS EC2, Vapi, ElevenLabs, Google cloud, Cloudways

REFERENCES

- Dr. Pratheeba Jeyananthan, Head of the department, Dept. of Computer Engineering Faculty of Engineering, University of Jaffna, pratheeba@eng.jfn.ac.lk, +94778448679
- Hirusha Deshan, Lead Software Engineer, Idea8 Pvt Ltd, Kottawa. hirusha@idea8.solutions, +94775306279
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